Wendy Q. Sun

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EDUCATION

Massachusetts Institute of Technology — Cambridge, MA
Double major: B.S. in Physics — 2022 – 2026
B.S. in Artificial Intelligence & Decision Making
Minor in Mathematics
GPA : 4.9/5.0

HONORS

Phi Beta Kappa Honor Society (top 10% of the graduating class) 2026
Sigma Pi Sigma Physics and Astronomy Honor Society 2026
Tau Beta Pi Engineering Honor Society 2026

PUBLICATIONS

3 primary author papers, ADS library. 16 total papers, ADS library.
† marks equal contribution.

Primary Author Papers

3. **Wendy Q. Sun**, Rohan P. Naidu, Hanpu Liu, Anna de Graaff, Jenny E. Greene, Jorryt Matthee, et al., *Overmassive No More: The Case for Little Red Dots Hosting Black Hole Seeds as Massive as Single Supermassive Stars*, submitted to OJAp [8 citations].
2. **Wendy Q. Sun**, Rohan P. Naidu, Jorryt Matthee, Anna de Graaff, et al., *Little Red Dot − Host Galaxy = Black Hole Star: A Gas-Enshrouded Heart at the Center of Every Little Red Dot*, OJAp, 9, 62505, 2026 [87 citations].
1. C. Loughridge†, **Q. Sun**†, et al., *DafnyBench: A Benchmark for Formal Software Verification*, The 4th Workshop on Mathematical Reasoning and AI at NeurIPS, 2024 [113 citations].

Contributing Author Papers

13. G. Limberg et al., *Chemodynamical Analysis of a CNO-Enhanced Ultra Metal-poor Star ([Fe/H] < −4): Insights into Early Enrichment by Faint Population III Supernovae*, submitted to AAS journals.
12. C. Ashall et al., *Can Little Red Dots Contribute To The Early Universe Cosmic Dust Budget?*, submitted to ApJ.
11. R. P. Naidu et al., *A gas-enshrouded and gas-reddened black hole at cosmic dawn*, Nature, 656, 329, 2026.
10. Y. Ishikawa et al., *Spatial decomposition of Little Red Dots with JWST/NIRSpec IFU into broad-line red cores and narrow-line blue host galaxies*, arXiv.
9. R. P. Naidu et al., *Little Red Dots as Intermediate Mass, Super-Eddington Engines: Insights from Type IIn Supernovae and The 1837–1856 Great Eruption of η Carinae*, submitted to OJAp.
8. A. Weibel, R. P. Naidu, et al., *Black Hole Stars Across the Universe: Identifying Central Engine Dominated Little Red Dots at $z \sim 1.5 - 9.5$*, submitted to OJAp.
7. B. Wang et al., *Water absorption confirms cool atmospheres in two little red dots*, arXiv.
6. A. Torralba et al., *A Black Hole Star at Cosmic Noon: Extreme Balmer break, photospheric continuum, and broad absorption by thick winds in a Little Red Dot at $z = 1.7$*, ApJL, 1005, 37, 2026.
5. Z. Liu, R. P. Naidu, A. Secunda, et al., *How I Wonder What You Are – JWST's Little Red Dots do not TWINKLE*, submitted to ApJ.

4. T. Li, **Q. Sun**, et al., *Fractal Generative Models*, Transactions of Machine Learning Research (TMLR), 2025.
3. J. Colaço Carr, **Q. Sun**, et al., *Focused Skill Discovery: Learning to Control Specific State Variables while Minimizing Side Effects*, The 2nd Reinforcement Learning Conference (RLC), 2025.
2. S. Casper, C. Ezell, et al., *Black-Box Access is Insufficient for Rigorous AI Audits*, The ACM Conference on Fairness, Accountability, and Transparency (FAccT), 2024.
1. W. Gurnee et al., *Universal Neurons in GPT2 Language Models*, Transactions of Machine Learning Research (TMLR), 2024.

PRESENTATIONS

ISSI Bern, *LRD − Host Galaxy = BH*: A Gas-Enshrouded Heart at the Center of Every LRD*
Meeting, Feb. 2026

RESEARCH EXPERIENCE

Weighing Central Engines of Little Red Dots — Honolulu, HI
Institute for Astronomy, University of Hawai'i at Mānoa • *Advisor: Rohan Naidu* — July 2026 – present
· Estimated the masses of central engines in little red dots by fitting stellar atmosphere models, finding masses consistent with a supermassive star origin.

Host Galaxy Separation for Little Red Dots — Cambridge, MA
MIT Kavli Institute • *Advisor: Rohan Naidu* — July 2025 – May 2026
· Separated the spectra of 90+ little red dots into central engine and host galaxy components, finding population-level evidence that their central engines are black hole stars.

Fractal Generative Models — Cambridge, MA
MIT CSAIL, He Vision Group • *Advisors: Tianhong Li, Kaiming He* — Nov. 2024 – Feb. 2025
· Contributed to developing and testing a novel AI generative model with fractal-like structure that nests smaller generative models within larger ones.

DafnyBench: A Benchmark for Formal Software Verification — Cambridge, MA
NSF IAIFI, Tegmark AI Safety Group • *Advisor: Max Tegmark* — Feb. – Nov. 2024
· Built the then-largest software verification benchmark (DafnyBench, 750+ programs) for evaluating how well AI models can verify code beyond just generating it.
· Built a VS Code extension that uses AI models to verify code (Dafny Autopilot, 400+ installs).

Focused Skill Discovery for Reinforcement Learning — Berkeley, CA
Center for Human-Compatible AI, UC Berkeley • *Advisor: Cam Allen* — June – Aug. 2024
· Implemented and evaluated algorithms that keep AI agents from causing harmful side effects.

SKILLS

Astronomy software Cloudy, Prospector, dynesty

RELEVANT COURSEWORK

Physics Astrophysics I (grad), General Relativity (grad), Statistical Mechanics I (grad), Relativistic Quantum Field Theory I (grad), Electromagnetic Theory (grad)
Mathematics Real Analysis, Numerical Analysis, Theory of Computation, Combinatorial Optimization